

AUTOMOTIVE ELECTRONICS

Industry Growth Prospects And Strategies

From electric vehicles (EVs) to autonomous vehicles, discover the challenges, key players, and market segments driving a \$136 billion industry revolutionising vehicles and the transportation landscape



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In the fast-paced world of automotive electronics, where innovation is the name of the game, businesses need to stay ahead of the curve to remain competitive. This exploration delves into the intricate world of automotive electronics, shedding light on key trends, challenges, and opportunities shaping the industry. The focus here centres on two critical components: sensors and power electronics, both of which play a pivotal role in the development of modern vehicles.

Northern Europe catching up with the trend. This shift towards EVs brings with it a unique set of challenges, particularly concerning battery management. To ensure the efficient operation and safety of EVs, many sensors are required to monitor factors such as battery charge, health, and temperature.

The industry is experiencing a growing demand for connected and software-defined vehicles. Features like vehicle-to-everything (V2X) communication, vehicle-to-grid (V2G), and vehicle-to-infrastructure (V2I) are becoming increasingly essential. These developments have led to a surge in demand for various electronic components to support these advanced features.

Increasing usage of electronics. The adoption of EVs and the integration of advanced features in modern vehicles have led to a substantial increase in the usage of electronic components in each automobile.

Growing number of connected vehicles. As vehicles become more connected, the demand for sensors and electronic

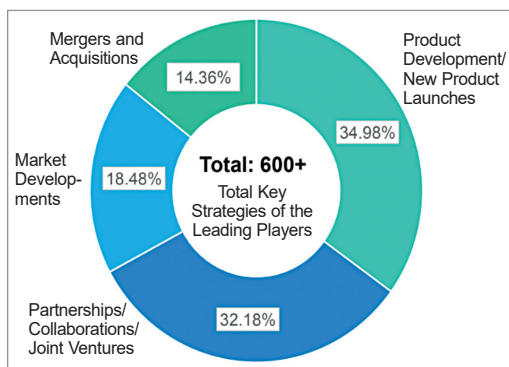


Fig. 1: Percentage of key strategies adopted

Prominent trends and drivers of growth

The automotive electronics industry has undergone a remarkable transformation in recent years. It has witnessed a surge in demand for electric vehicles (EVs), with regions like India and

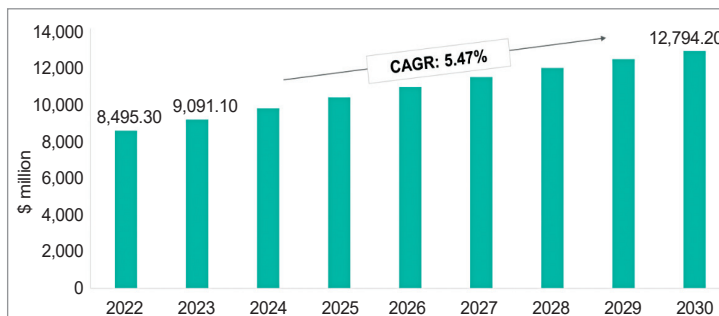


Fig. 2: Global conventional sensor market scenario—present and projected

